



Detection of parkinson's disease with neuroimaging modalities using machine learning and artificial intelligence: a systematic review

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Received: 14 August 2025 / Accepted: 16 December 2025 / Published online: 14 January 2026
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Abstract

The application of machine learning (ML) and artificial intelligence (AI) algorithms in medical imaging is an emerging area of interest, particularly in the context of clinical decision-making. Here, we report on the overall performance (i.e., sensitivity, specificity, and accuracy) of commonly used ML/AI techniques including convolutional neural networks (CNNs), support vector machines (SVMs), random forests, and ensemble approaches on the clinically relevant task of distinguishing between Parkinson's disease (PD) participants and matched healthy controls (HC). Our systematic review includes 130 studies from six different imaging modalities – dopamine transporter scans ($[^{123}\text{I}]$ Ioflupane single-photon emission computed tomography (SPECT)), positron emission tomography (PET) including $[^{18}\text{F}]$ FDG, $[^{18}\text{F}]$ DOPA, and $[^{11}\text{C}]$ raclopride, structural magnetic resonance imaging (MRI) (T1- and T2-weighted), functional MRI, and diffusion MRI. While some findings were in line with expectations for some modalities, such as the superior performance of dopamine SPECT and PET (>90% sensitivity, specificity, and accuracy with methods like convolutional neural networks), others were more nuanced with the best-performing class of algorithms depending on the imaging modality, and sometimes, even the data source. Overall, we summarize the emerging trends across studies for each imaging technique and provide valuable recommendations for future lines of inquiry.

Keywords Parkinson's disease · Artificial Intelligence · PET · MRI · SPECT

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Introduction

With an age-standardized incidence rate of 13.43 per 100,000 in 2019, Parkinson's disease (PD) is the second most diagnosed neurodegenerative condition worldwide [1]. Not only does this age-related progressive disease lead to severe motor disability in patients, but it also has detrimental effects on the cognitive and psychiatric state of the affected individual while being an economic burden on caregivers [2]. Previous research suggests that the adverse consequences of PD progress with age, creating a cycle of increased physical, emotional, cognitive, and socio-economic costs [2]. Therefore, timely diagnosis of PD with the help of neuroimaging and similar non-invasive diagnostic methods is of paramount importance, preferably in the pre-clinical or early clinical phase of the disease [3].

Aside from a timely diagnosis, another challenge with movement disorders is the possibility of misdiagnosing a patient due to the similarity between the symptoms of PD

and other Parkinsonian syndromes [4–7]. It is noteworthy that the treatment of these motor disorders differ substantially and misdiagnoses result in patients being deprived of the correct standard of care in the critical early stages of the disease, when treatment is generally more effective in managing symptoms [8]. Machine learning (ML) methods, including artificial intelligence models (AI), have demonstrated tremendous potential with regard to medical diagnostics [9]. AI is also expected to lead to a paradigm shift in the realm of medical imaging [9], with the impact being particularly enhanced in areas where timely diagnosis and early management can significantly improve outcomes [10]. This can bridge the gap between imaging abnormalities and quantified data systems [11, 12], resulting in faster, more accurate, and more efficient decision-making [13, 14]. Moreover, such systems are capable of integrating multi-modal clinical and imaging data to help physicians and patients reach a more comprehensive summary, which further amplifies their ability to treat diseases, manage disorders, and plan follow-up care [15]. A recent study that differentiates PD, Multiple System Atrophy (MSA), and Progressive Supranuclear Palsy (PSP) using a support vector machine on diffusion MRI sequences in a large multi-center cohort is an excellent example of the adaptation of machine learning and AI techniques for clinical diagnosis. They reported high diagnostic accuracy (96%), with autopsy-validated precision (94%), outperforming clinical diagnosis and demonstrating robustness across scanner types [16].

Numerous previous studies have investigated the applications of AI in PD diagnosis, with AI models consistently demonstrating promising results. However, there is a lack of systematic reviews of the performance of these methods in PD. Such insights are important not only for comparing the performance of several contemporary data-driven methods across different datasets and modalities, but also for understanding potential variability arising from different implementations of the same underlying methods and datasets. This study presents a systematic review of the performance of ML and AI methods, solely in the diagnosis and discrimination of the PD from controls. By keeping the focus squarely on studies that discriminate PD from controls, we aim to synthesize the reported diagnostic performance of ML and AI methods in discriminating PD from controls using neuroimaging data.

Methods

We conducted a systematic review following the Prisma 2020 guideline [17]. The protocol for this review was registered at PROSPERO (CRD42023438000). We assessed existing research that employed machine learning and deep learning

to detect and classify PD from Healthy Control (HC) using neuroimaging. All data extracted are from published articles, so no ethical approval and patient consent were needed.

Search strategy

In July 2023, seven biomedical databases were searched to retrieve the screening citations: Ovid MEDLINE(R) ALL, Web of Science-Core Collection, Scopus, Cochrane CENTRAL, Embase, PubMed, and APA PsycINFO. In addition, Institute of Electrical and Electronics Engineers (IEEE) (<https://www.ieee.org/>) was also searched. The searches comprised of three themes: neuroimaging, artificial intelligence, and PD. An example of the search in MEDLINE can be found in Table S1. The searches were peer-reviewed by another librarian using the Peer Review of Electronic Search Strategies form [18, 19]. An updated search was done using the same keywords and databases, and therefore, the search for the systematic review presented included all articles until May 2024 (Fig. 1).

The inclusion and exclusion criteria are summarized below:

Inclusion criteria.

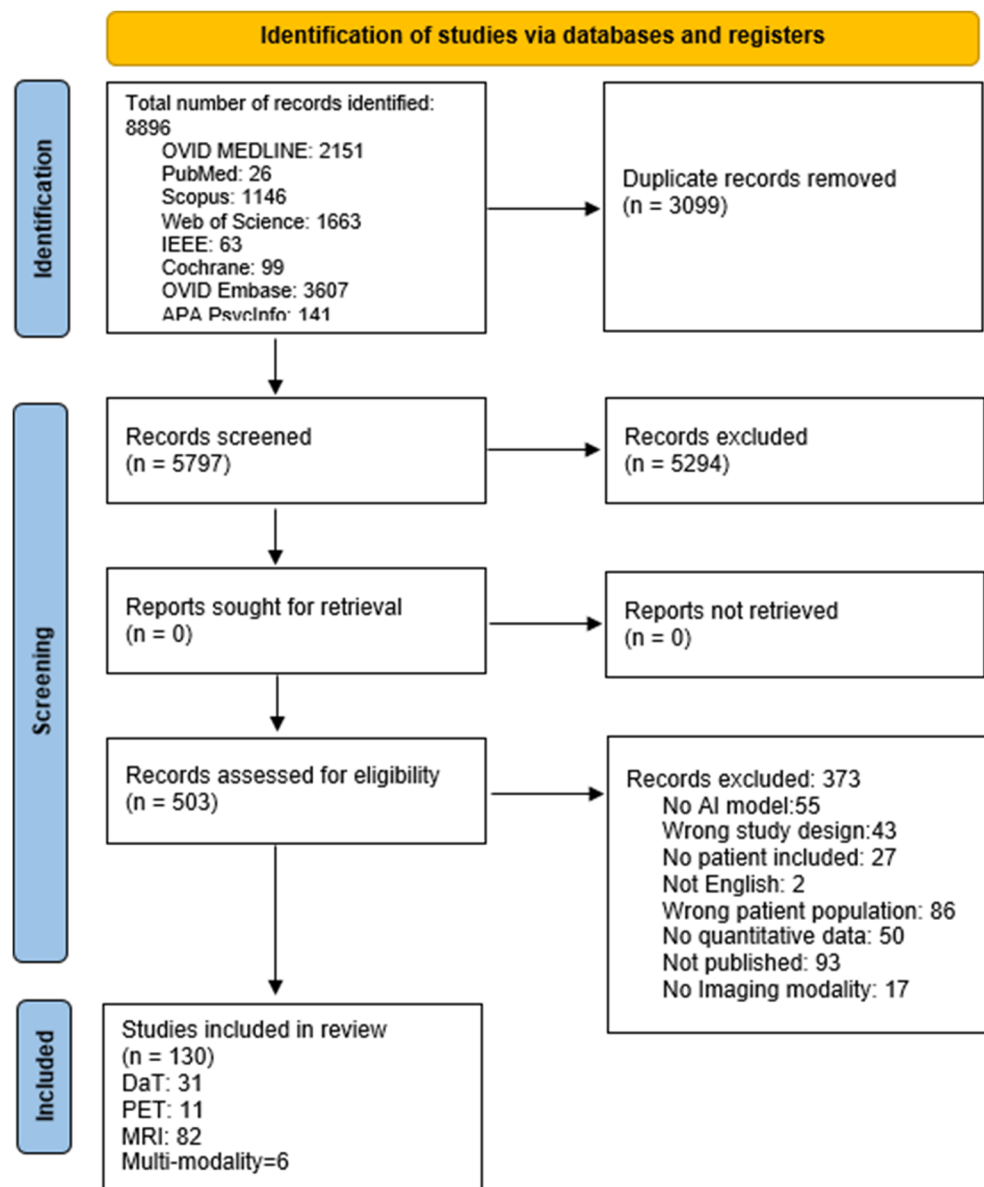
1. Retrospective and prospective articles written in English that utilized ML or deep learning models on dopamine transporter scans (DaT scan or [^{123}I] Ioflupane single-photon emission computed tomography (SPECT)), dopamine positron emission tomography (PET), [^{18}F] FDG PET, or magnetic resonance imaging (MRI) for classifying PD.
2. Studies reporting diagnostic accuracy, sensitivity, and specificity metrics for PD detection.
3. Papers including both PD patients and HCs.

Exclusion criteria.

1. Review papers, meta-analysis papers, case reports, case series papers, and conference papers.
2. Studies involving participants with other neurological disorders or lacking HCs, as well as studies with less than 5 participants.
3. The absence of ML or AI methods.
4. Studies that did not report the statistical metrics.

Screening and data extraction

Search results were pooled in EndNote and de-duplicated (www.endnote.com). This set was uploaded to Covidence (www.covidence.org) for screening. At least two authors (SH, SB, SFN, SC) participated in the screening of papers in two stages: first, title and abstract screening, and then

Fig. 1 Flow diagram showing the inclusion and exclusion strategy

full-text screening. In cases of disagreement, a separate reviewer made the decision (FE). Data was extracted by two different reviewers and encompassed three key domains:

- 1) Study characteristics (first author, year of publication, data source) (Supplementary Table 2).
- 2) Participants information (number of PD and HC subjects, age reported as mean (SD)).
- 3) Imaging modality, ML or AI model, and related metrics (e.g., accuracy, specificity, sensitivity).

In cases where multiple models/features of classification performance metrics were reported in the studies, all measures were initially extracted. If a study reported results from multiple ML models (e.g., using different ML methods and sets of features), the highest accuracy is reported.

Quality assessment of included studies was performed using eight domains adapted from the PRISMA 2020 guidelines and the QUADAS-2 framework for diagnostic accuracy studies [20]. Each domain was rated as “Yes” (Y), “No” (N), “Unclear” (U), or “Not Reported” (NR). Full results of the quality assessment are summarized by modality in the Supplementary Table 5.

Results

Literature review

Our systematic review included 130 articles in total, including 82 MRI, 31 DaT scan, 11 PET, and 6 multimodal studies. The first publication was in 2009.

Data source and characteristics of included studies

For the study design, we categorized the articles into two main groups: Parkinson's Progression Markers Initiative (PPMI) and non-PPMI datasets. Approximately 54% of the articles are in the PPMI group, 33% in the Non-PPMI group, around 3% use multiple datasets, and 9.2% did not specify their dataset. The median sample size was 239. A summary of the demographic information for all included studies is provided in Table S2.

Brief background for specific ML/AI models

This paper summarizes the overall performance of many different ML/AI models across 130 studies, each with its own implementation of the methods that could include variability in terms of tuned parameters, hyperparameters, and modeling choices. Therefore, a complete explanation of even the general classes of models utilized across all studies is outside the scope of this paper, and we rely on the familiarity of the readers with ML/AI, classical regression, and data-driven multivariate methods. However, brief explanations of a selected, small set of methods most relevant to the results section are provided below, along with references for further reading.

1. **Support Vector Machine (SVM):** SVM is a supervised machine learning model (i.e., it needs ground-truth labels for training data) that finds an optimal hyperplane that best separates data points by maximizing the margin between them in a higher-dimensional space (for brain images, data from many regions or voxels can be considered a high-dimensional dataset). Further reading about the theory and use of many different variants of the classic SVM methods in medical imaging can be found here [21].
2. **Logistic Regression and LASSO:** Logistic regression models the posterior probability of classification. This provides a more nuanced classifier which, in addition to automatic classification, also informs about the certainty (i.e. probability) of that classification. For high dimensional data, such as images, sparse logistic regression, called LASSO, is commonly used. Additional information about Logistic Regression and Lasso can be found in [22].
3. **Neural Networks:** A neural network is an artificial intelligence model whose training could be supervised or unsupervised, consisting of interconnected layers of nodes that process and transform input data. The overall transformation contains non-linear elements. It works by minimizing an overall loss of function and using this to adjust the weights of the connections between nodes. Depending on the network architecture (number of layers of nodes) and the topology of connections

between them, neural networks are further subclassified into many different categories. Neural networks can be designed for classification or regression problems and have been extensively used in medical imaging [23].

4. **Convolutional Neural Networks (CNNs):** A CNN is a specific type of neural network that uses convolutional layers that apply linear filters to local areas within an image, followed by pooling and fully connected layers to classify or regress data. This type of network is especially useful for extracting local features from an image, like edges or texture, and then using them as the basis for final predictions. We report results from many sub-categories of CNNs. Some of these indicate legacy architecture and pooling function choices (such as Alex NET) [24]. In contrast, others use architectures with deep hidden layers (Visual Geometry Group or VGG) [25] or architectures where layers are connected to every other layer in a feed-forward fashion (Dense NET) [26]. A review of deep CNNs in neuroimaging with MRI can be found here [27].

Apart from these, the studies have also made use of several classical machine learning methods such as methods to reduce data-dimensionality and complexity such as principal component analysis and linear discriminant analysis; clustering approaches such as k-nearest means and hierarchical clustering.

Performance metrics

Various metrics have been reported in different studies to assess the performance of ML/AI models. Here, we utilize sensitivity, specificity, and accuracy in classification between PD and controls as the main indicators of performance as these were the widely reported outcome measures. The mean and standard deviation (SD) for each modality is reported separately in their respective tables.

Performance of ML/AI models on different datasets

The imaging modalities included in the study are DaT imaging, which accounted for 23% of the studies; [^{18}F]FDG PET, making up 6%; and PET imaging of the dopamine system, utilized in 3% of the studies. Structural MRI was the most used modality at 43%, functional MRI (fMRI) included 9% of the studies, diffusion MRI represented 7% of the studies, and combined modality studies and specific types of MRI made up 5% and 4%, respectively.

DaT scan

31 studies used ML/AI models on DaT scan data. The most common models used were 2D and 3D CNNs, followed

by SVM and a few other techniques like linear regression or hierarchical clustering. Overall, accuracy was $94.9\% \pm 7.1\%$ (64% to 100%). Also, the sensitivity and specificity measures were $94.9\% \pm 5.9\%$ (78% to 100%) and $92.3\% \pm 12.9\%$ (38% to 100%), respectively. Detailed results are presented in Table 1.

PET imaging

A) [^{18}F]FDG PET

Among the 11 PET studies analyzed, 8 utilized [^{18}F]FDG PET as the imaging modality. Support SVM and RF emerged as the most frequently applied classification models. Overall accuracy was $83.8\% \pm 33.4\%$ (70%–97.1%). The sensitivity was $83.8\% \pm 20.7\%$ (39%–98%), while the specificity was $66.5\% \pm 43.3\%$ (10%–98%). Detailed results are provided in Table 2.

B) [^{18}F]DOPA and [^{11}C]raclopride PET

Four studies in this group utilized two distinct dopamine-based radiotracers, [^{18}F]DOPA and [^{11}C]raclopride. The accuracy across these studies was $90.1\% \pm 6.7\%$ (81%–96%), with a sensitivity $97.1\% \pm 3.5\%$ (93.2%–100%) and a specificity of $84.4\% \pm 18.3\%$ (64.3%–100%). Detailed results are provided in Table 3.

MRI

Different types of MRI techniques, including structural MRI, functional MRI, diffusion-weighted imaging (DWI), diffusion tensor imaging (DTI), multimodal MRI, and other specialized MRI methods, were employed to differentiate PD from HC. Results for the three primary MRI modalities are presented in the main manuscript, while outcomes for multimodal and specialized MRI techniques are provided in the supplementary materials (Tables S3 and S4).

A) Structural MRI (sMRI)

A total of 56 articles applied AI models to structural MRI (sMRI), including T1- or T2-weighted sequences, with 21% utilizing Non-PPMI dataset. SVM was the most commonly used model, followed by CNN. The reported accuracy was $84.3\% \pm 13.4\%$ (49.6%–100%), with a sensitivity of $85\% \pm 18.7\%$ (65.2%–100%) and a specificity of $82.7\% \pm 17.2\%$ (45.5%–100%). Detailed results are provided in Table 4.

B) Functional MRI (fMRI)

A total of 12 articles applied AI models to fMRI, all utilizing resting-state fMRI data. Of these, 66% of the studies

used non-PPMI datasets. SVM was the most frequently used model. The accuracy across studies was $85.4\% \pm 9.1\%$ (67.4%–100%), with a sensitivity of $79.4\% \pm 7\%$ (70.3%–90.9%) and a specificity of $82.8\% \pm 11.4\%$ (62.5%–94%). Detailed results are presented in Table 5.

C) Diffusion MRI (dMRI)

Nine articles utilized diffusion imaging techniques, including diffusion tensor imaging (DTI) and diffusion-weighted imaging (DWI), with DTI being the most commonly employed. Among these, 33% used a Non-PPMI dataset. CNN and SVM were the most frequently applied models. The accuracy reported was $79.1\% \pm 10.6\%$ (58%–92.4%), with a sensitivity of $73.2\% \pm 15.3\%$ (56%–92.6%) and a specificity of $72.1\% \pm 23.9\%$ (41%–91.7%). Detailed results are reported in Table 6.

Discussion

In this systematic review, we explored 130 published studies from 7 databases employing ML/AI algorithms for detection of PD from HC. We included studies from all major types of imaging modalities, including DaT SPECT, PET imaging of the dopamine system, [^{18}F]FDG PET, and structural, functional, and diffusion MRI. Since this represents a vast space across modalities and ML/AI methods employed, we purposely limited our focus on overall generalizability of group discrimination of these methods across studies with different cohorts and techniques. We believe that this is an important first step in generating useful insights for the development of computation assisted detection which could guide next steps or identify areas requiring more research. Our review was not designed to statistically compare model performance across studies, given variability in sample sizes, diagnostic criteria, input modalities, and reporting practices. Instead, we aimed to summarize overall trends to inform future model development and standardization.

First, it is important to note that the overall performance of ML/AI methods was best for datasets based on DaT scans or PET imaging of the dopaminergic system. Deep learning techniques like CNNs (including architectures such as Visual Geometric Group [VGG] and AlexNET) consistently outperformed all other methods with very high sensitivity, specificity, and accuracy (~90–95% for all three on average). SVM was another algorithm that showed promise although its performance was more variable with some implementations showing lower accuracy of classification (80–85%). Since PD causes dopaminergic neuronal loss in the basal ganglia, the high performance of data-driven methods in DaT SPECT and [^{18}F]DOPA and [^{11}C]raclopride PET.

Table 1 ML/AI models and extracted metrics for included DaT scan articles

First author	Machine learning model	Radiotracer	Outcome measure	Reference region	Sensitivity	Specificity	Accuracy
Chiu et al. [28]	ml-mt-vs	Tc-99 m TRODAT-1	SBR	Occipital cortex	87	70	84
Yoon et al. [29]	XGBoost	¹⁸ F-FP-CIT	SUV				96
Magesh et al. [30]	CNN with VGG16	¹²³ I-FP-CIT	Preprocessed Images		97.5	90.9	95.2
Prashanth et al. [31]	SVM	¹²³ I-FP-CIT	SBR	Occipital cortex	97.03	95.01	96.4
Bouchrouite et al. [32]	SVM	¹²³ I-FP-CIT	SBR	Occipital cortex	95.3	98	97.3
Castillo-Barnes et al. [33]	WMV	¹²³ I-FP-CIT	Intensity pattern		92.5	95	93.8
Bhalchandra et al. [34]	SVM	¹²³ I-FP-CIT	SBR	Occipital cortex			99.4
Khachnaoui et al. [35]	Hierarchical Clustering	¹²³ I-FP-CIT	SBR	Occipital cortex	78.1	38.8	64
Kurmi et al. [36]	VGG16	¹²³ I-FP-CIT	Preprocessed Images		98.8	97.6	98.4
Manimegalai et al. [37]	VGG16	¹²³ I-FP-CIT	Preprocessed Images				97
Martinez-Murcia et al. [38]	ALEXNET	¹²³ I-FP-CIT	Preprocessed Images		93.3	95.8	94.1
Oliveira et al. [39]	SVM	¹²³ I-FP-CIT	BP _{ND}	Cortex VOI	97.7	98	97.8
Oliveira et al. [40]	SVM	¹²³ I-FP-CIT	SBR	Whole nonspecific brain region	98	97.6	97.9
Pereira et al. [41]	CNN	¹²³ I-FP-CIT	Preprocessed Images		92.9	91.9	92.4
Pianpanit et al. [42]	Deep PD net	¹²³ I-FP-CIT	SBR	Occipital cortex	97.1	95.4	96.7
Prashanth et al. [43]	SVM	¹²³ I-FP-CIT	Shape-Based Features & SBR	occipital cortex for SBR	97.3	97.1	97.2
Shiiba et al. [44]	SVM	¹²³ I-FP-CIT	Shape-Based Features & SBR	occipital cortex for SBR	98	95	
Shiiba et al. [45]	Radiomics-Based ML	¹²³ I-FP-CIT	SBR	Occipital cortex	100	85	96.8
Thakur et al. [46]	CNN	¹²³ I-FP-CIT	Preprocessed Images		99.2	99.4	99.2
Wenzel et al. [47]	CNN	¹²³ I-FP-CIT	SBR	Whole brain	98.3	96.2	97.2
Zhou et al. [48]	LR	¹²³ I-FP-CIT	SBR	Self-normalization			98.9
Choi et al. [49]	Deep CNN	¹²³ I-FP-CIT	SBR	Occipital cortex	94.2	96	100
Dehghan et al. [50]	CNN	¹²³ I-FP-CIT	Preprocessed Images		98.8	97.8	98.5
Ortiz et al. [51]	CNN	¹²³ I-FP-CIT	Isosurfaces-Based Feature Extraction		95	95	95
Quan et al. [52]	Deep CNN	¹²³ I-FP-CIT	Preprocessed Images		98	97	98
Castillo-Barnes et al. [53]	SVM	¹²³ I-FP-CIT	Morphological & Voxel Features		95.4	97.4	96.4
Bollipo et al. [54]	Superior vector regression	¹²³ I-FP-CIT	SBR	Occipital cortex			93.5
Siluvaimuthu et al. [55]	ANN	¹²³ I-FP-CIT	Preprocessed Images				100
Khachnaoui et al. [56]	CNN	¹²³ I-FP-CIT	SBR based on image slices				98.4
Hsu et al. [57]	SVM	Tc-99 m TRODAT-1	Dopamine Transporter Activity Volume & Maximum		81.8	100	82.2
Tagare et al. [58]	Logistic Lasso, Logistic PCA	¹²³ I-FP-CIT	SBR	Occipital cortex	97.2	96.2	96.5
Mean					94.9	92.3	94.9
SD					5.4	12.9	7.1

Abbreviations: *ml-mt-vs* machine learning-visual scale, *xgboost* extreme gradient Boosting, *CNN* Convolutional neural Networks, *VGG* Visual geometry Group, *SVM* Support vector Machine, *WMV* Weighted majority Voting, *LR* Logistic regression, *ANN* Artificial neural Network, *SBR* striatal binding ratio, *BP_{ND}* binding Potential, *VOI* Volume of Interest, articles that didn't use PPMI dataset are highlighted in Gray in all tables

Table 2 ML/AI models and extracted metrics for [¹⁸F]-FDG PET studies

First author	Tracer	Model	Sensitivity	Specificity	Accuracy
Glaab et al. [59]	[¹⁸ F]-FDG	RF	94.1	12.5	74.6
Shen et al. [60]	[¹⁸ F]-FDG	Deep Belief Network	96	84	90
Manzanera et al. [61]	[¹⁸ F]-FDG	SVM	39	98	70
Sun et al. [62]	[¹⁸ F]-FDG	Deep learning	88.9	97.8	95.1
Veen et al. [63]	[¹⁸ F]-FDG	LVQ	79.2	68.1	
Wu et al. [64]	[¹⁸ F]-FDG	SVM	92	82	90.1
Junck et al. [65]	[¹⁸ F]-FDG	RF	98	10	76
Dai et al. [66]	[¹⁸ F]-FDG	DenseNet		97.5	97.1
Mean			83.8	66.5	83.8
SD			20.7	43.3	33.4

Abbreviations: *RF* Random Forests, *SVM* Support Vector Machine, *LVQ* Learning Vector Quantization**Table 3** ML/AI models and extracted metrics for other PET studies

First author	Tracer	Model	Sensitivity	Specificity	Accuracy
Glaab et al. [59]	[¹⁸ F]-FDOPA	SVM	93.2	64.3	81
Iep et al. [67]	[¹⁸ F]-FDOPA	K-NN			89.9
Martins et al. [68]	[¹¹ C] raclopride	ML	98.1	100	96.7
Piccardo et al. [69]	[¹⁸ F]-FDOPA	CNN	100	89	93
Mean			97.1	84.4	90.1
SD			3.5	18.3	6.7

Abbreviations: *SVM* Support Vector Machine, *K-NN* k-nearest neighbors, *ML* Machine Learning, *CNN* Convolutional Neural Network

to discriminate subjects with clinical PD from controls is unsurprising. It should be noted that while dopamine PET tracers demonstrated consistently high sensitivity ($97.1 \pm 3.5\%$), their specificity was more variable and, on average, lower ($84.4 \pm 18.3\%$). This pattern suggests that these models are more effective at detecting PD cases than excluding non-PD, which could increase the risk of false positives. From a clinical standpoint, this limitation highlights the need for caution in interpreting dopamine PET models in isolation and supports the integration of multi-modal imaging or clinical markers to improve diagnostic specificity. However, the consistently high performance of CNNs in this task suggests that this method is probably the most repeatable and, therefore, an ideal candidate for early identification of PD patients from controls in early stages of the disease where visual readout by experts may be more challenging. Prospective studies should apply CNNs to DaT scans from at-risk cohorts at an early stage (such as those with REM sleep behavior disorder or individuals at risk through either genetic factors or identified through seed amplification assays for α -synuclein) to test whether it can accurately discriminate individuals that develop PD. A major limitation for studies reported under this modality was the overutilization of the PPMI DaT SPECT dataset (29/31 studies used PPMI), which might raise concern over the generalizability of these findings. However, given the relatively higher signal change in dopamine imaging due to PD which can be harnessed by these AI models, the superior performance of CNNs over other algorithms on the same dataset may even be considered indicative of the robustness

of this technique. Nevertheless, we recommend that the trained CNN and SVM models should be tested in independent datasets, possibly with additional variability in cohort demographics, such as cohorts like Parkinson's disease and Alzheimer's disease Neuroimaging Initiative (PADNI) dataset [144].

For DaT SPECT and PET, the voxel values are often normalized to a region where the tracer binding is predominantly non-specific, such as the occipital cortex or cerebellum (or some parts of these regions) and this could contribute to an independent source of variability across studies [58]. While the majority of the dopamine SPECT and PET studies included here did utilize some reference regions (predominantly occipital cortex), it should be noted that methods like CNNs can be trained over the entirety of dose-normalized images, without further normalization to reference regions, and this could have benefits of better selection of voxels which can provide information about non-specific binding, compared to a pre-specified reference. In the same vein, it is important that relevant dopaminergic voxels outside of the caudate and putamen, particularly within the thalamus and globus pallidum for human studies, are included in the analyses [58]. Particularly, for DaT SPECT, performance may be improved by utilizing the entire 3D images, compared to specific 2D image slices or maximum intensity projections.

There is another important caveat to the above discussion. Machine learning and AI classifiers implicitly assume a mathematical feature space in which to operate. Classification performance improves significantly when the right feature space for a problem is discovered and used.

Table 4 Represent the AI models and extracted metrics for included structural MRI articles

First author	MRI type	machine learning model	Sensitivity	Specificity	Accuracy
Salvatore et al. [70]	T1W	SVM	84.5	81.9	83.2
Focke et al. [71]	T1W	SVM	33.3	45.4	39.5
Wang et al. [72]	T1W	CNN	77.1	80.6	78.7
Kiryu et al. [73]	T1W	CNN	94.4	97.8	96.8
Lampe et al. [74]	T1W	DNN			49.6
Lee et al. [75]	T1W	Structural covariance networks	81	69	75
Tupe-Waghmare et al. [76]	T1W	Radiomics			70
Rana et al. [77]	T1W	SVM	90	86.6	88.3
Duchesne et al. [78]	T1W	SVM	93	88	91
Huppertz et al. [79]	T1W	SVM	65.2	67.1	65.7
Gore et al. [80]	T1W	SVM			83.3
Cigdem et al. [81]	T1W	SVM	95	92.5	93.7
Peng et al. [82]	T1W	SVM	87.6	87.8	85.8
Singh et al. [83]	T1W	SVM	96.9	84.1	90.8
Sun et al. [84]	T1W	Radiomics	100	100	100
Amoroso et al. [85]	T1W	SVM	93	92	93
Yang et al. [86]	T1W	ResNet	95.2	97.4	96.1
Singh et al. [87]	T1W	SVM			93.2
Chakraborty et al. [88]	T1W	SVM			95.3
Chakraborty et al. [88]	T1W	CNN			95.2
Chattopadhyay et al. [89]	T1W	CNN			49
Dhinagar et al. [90]	T1W	CNN			70.6
Tafari et al. [91]	T1W	Radiomics	75.4	68	71
Ganotra et al. [92]	T1W	SVM	82	76.5	80
Pahuja et al. [93]	T1 W	ELM	98.7	98.7	97.9
Song et al. [94]	T1W	ReliefF	89.6	80	84.9
Zeng et al. [95]	T1W	SVM			97.7
Erdas et al. [96]	T1W	CNN			96
Camacho et al. [97]	T1W	CNN	7.7	81.3	79
Li et al. [98]	T1W	SVM	87.1	100	93.5
Babu et al. [99]	T1 W	McRBFN	83.3	81	82.3
Balchandra et al. [100]	T1W	SVM			78
Lavalle et al. [101]	T1W	SVM	99.3	87.5	95
Yang et al. [102]	T1W	KNN			87.1
Sivaranjini et al. [103]	T1W	Radiomics			72.4
Kaur et al. [104]	T1W	Deep CNN	89	90.2	89.2
Shi et al. [105]	T1W	RVFL	74.2	77	75.6
Esmailzadeh et al. [106]	T1W	CNN			72
Gong et al. [107]	T1W	DNN	82.9	80	81.7
Adeli et al. [108]	T1W	LDA	84		81
Praneeth et al. [109]	T1W	CNN	98.1	96.8	98.8
Begum et al. [110]	T1W	DCNN			94
Kanagaraj et al. [111]	T1W	DenseNet	99.6		99.9
Sailaja et al. [112]	T1W	ResNet			100
Yasaka et al. [113]	T1W	Intracellular volume fraction	79	69	74
Chen et al. [114]	T1W	SVM	97.4	90.4	95
Cui et al. [115]	T2W	DL	92	87	90
Peng et al. [116]	T2W	KPR+	98	13.3	66.2
Vyas et al. [117]	T2W	CNN			88.9
Sivaranjini et al. [118]	T2W	AlexNET	89.3	88.4	88.9
Yin et al. [119]	T2W	CNN			77.9
Liu et al. [120]	T2W	Radiomics			
Cui et al. [115]	T2W	MeshNet			76.9
Hussain et al. [121]	T2W	CNN	96.8	95.8	96
Akram et al. [122]	T1W/T2W	FBLstmNet		99.5	99.5

Table 4 (continued)

First author	MRI type	machine learning model	Sensitivity	Specificity	Accuracy
Veetil et al. [123]	T2W	VGG19			92.6
Mean			85	82.7	84.3
SD			18.7	17.2	13.4

Abbreviations: *SVM* Support Vector Machine, *CNN* Convolutional Neural Network, *DNN* Deep Neural Network, *LASSO* Least Absolute Shrinkage and Selection Operator, *ResNet* Residual Network, *ELM* Extreme Learning Machine, *McRBFN* meta-cognitive radial basis function network, *RVFL* Random Vector Functional Link Network, *LDA* linear discriminant analysis, *DL* Deep Learning, *KPR+* kernel ridge regression plus, *VGG* Visual Geometry Group

Table 5 Represent the AI models and extracted metrics for included fMRI articles

First author	MRI type	Outcome measure	Machine learning model	Sensitivity	Specificity	Accuracy
Tian et al. [124]	RS-fMRI	ALFF	SVM	73.6	86.5	80.7
Long et al. [125]	RS-fMRI	ALFF	SVM	78.9	92.5	86.9
Rubbert et al. [126]	RS-fMRI	Between-network functional connectivity	LR	81	72.7	76.2
Shi et al. [127]	RS-fMRI	ALFF	Radiomics	78.8	76	78
Shi et al. [128]	RS-fMRI	ALFF	LASSO	70.37	62.5	67.4
Wu et al. [129]	RS-fMRI	Functional connectivity	SVM			91.2
Islam et al. [130]	RS-fMRI	Resting-state networks	3D-CNN	87.4	85	86
Chen et al. [131]	RS-fMRI	Functional Connectome differences	LR	90.9	93.3	92.3
Xu et al. [132]	RS-fMRI	Functional connectivity	SVM			96
Dutta et al. [133]	RS-fMRI	Functional connectivity	Canonical Independent Component Analysis			87.5
Abós et al. [134]	RS-fMRI	Functional connectivity	SVM	74	94	82.6
Cao et al. [135]	RS-fMRI	Functional connectivity	SVM			100
Mean				79.4	82.8	85.4
SD				7.0	11.4	9.1

Abbreviations: *SVM* Support Vector Machine, *LR* Logistic Regression, *CNN* Convolutional Neural Network, *ALFF* Amplitude of Low-Frequency Fluctuations

Table 6 Represent the AI models and extracted metrics for included diffusion MRI articles

First author	MRI type	Outcome measure	Machine learning model	Sensitivity	Specificity	Accuracy
Tsai et al. [136]	DTI	FA, MD, AD, RD	SVM	76.2	65.6	76.1
Li et al. [137]	DTI	FA, MD	radiomics	87.5		
Zhao et al. [138]	DWI	FA, MD	CNN			78
Huang et al. [139]	DTI	L1, V1	Embed-ding learning	68.3	90.1	83.3
Yang et al. [102]	DTI	FA, MD	K-NN			76.2
Zhang et al. [140]	DTI	FA	RF			78
Lei et al. [141]	DTI	FA, MD	Joint regression			91.1
Prasuhn et al. [142]	DTI	FA	SVM	56	41	58.1
Zhao et al. [143]	DTI	FA	CNN	92.6	91.7	92.4
Mean				73.2	72.1	79.1
SD				15.3	23.9	10.6

Abbreviations: *SVM* Support Vector Machine, *CNN* Convolutional Neural Network, *K-NN* k-nearest neighbors, *RF* Random Forest, *FA* Fractional anisotropy, *MD* mean diffusivity, *AD* axial diffusivity, *RD* radial diffusivity, *L1* First eigenvalue, *V1* first eigenvector

Recent work has also identified nonlinear spaces similar to projective spaces for DaTscan image analysis [42]. Using such spaces, it is possible to use simpler classifiers such as LASSO and obtain classification performance (98% accuracy) similar to or better than CNN performance [42]. These new spaces make also make it possible to establish precise mathematical theorems that can explain how classification with and without the use of a non-specific binding reference region relates to each other.

In comparison to dopamine SPECT or PET, the overall accuracy and especially specificity of discrimination was lower for [¹⁸F]FDG PET (about 85% and 70%, respectively). It should be noted that we only had 8 studies however, so these findings should be interpreted with caution. Investigators of these studies did not use CNNs, instead preferring random forests, SVM and some other niche machine learning or deep learning approaches such as localized generalized matrix learning vector quantization (LGMLVQ)

and DenseNets (a type of CNN utilizing dense connections between layers), respectively. Methods incorporating deep learning with radiomics features and DenseNets showed high accuracy and specificity, though as these are singular studies, replication in other cohorts is desirable for claims of robustness. Another potential area of investigation should be the comparison of these methods to the multivariate method scaled sub profile model principal component analysis, or SSM-PCA [145]. SSM-PCA has been used in many PD studies to identify a highly reproducible metabolic pattern called Parkinson's related motor pattern or PDRP, and the expression strength of this pattern has been shown to discriminate PD from controls (see Supplementary Table S6). Though most [^{18}F]FDG PET studies choose to apply SSM-PCA to derive PDRP loadings for their own datasets (which uses labels of PD vs. controls), it should be noted that the topographical loadings for the PDRP can be averaged from these previously published studies and applied prospectively to perform classification of PD vs. controls. It would be highly desirable to compare the discriminative performance of ML/AI methods on [^{18}F]FDG PET (or equivalently on fMRI) data to the benchmark set by the prospective application of known PDRP loadings on the same dataset to probe the addition benefits of AI/ML against computational costs. However, it was encouraging to observe a much higher spread of non-PPMI data sources for ML/AI studies with [^{18}F]FDG PET, even though the overall number of studies remains small for this modality.

Across more than 50 structural MRI studies, the average accuracy, sensitivity, and specificity of group classification for ML/AI methods were lower than that seen with dopamine SPECT or PET (average between 80 and 85% for all three measures). While most studies utilized T1-weighted images, a handful of studies (about 20%) instead used T2-weighted images. However, we did not find any statistically significant effect of the MR-weighting on the accuracy of classification, with both weightings leading to a similar result, on average. Structural MRI studies reported the most diverse use of ML/AI algorithms with CNNs and their variants (such as DenseNets or deep residual CNNs), SVMs, and radiomics-based approaches being the most common choices, though other methods such as Gaussian processes, LASSO regression, linear discriminant analysis, and others were also tried. Unlike DaT SPECT, fluorodopa and raclopride PET, the results of CNN-based and SVM-based methods were highly variable across studies, with accuracies ranging from 70% to 100% for most studies. According to this high variability in performance, and especially for studies with very high accuracies, it would be useful to challenge these published algorithms, especially those with reported >85% accuracy, to larger and new datasets before algorithm deployment to truly understand which models

generalize best. Such an exercise should be possible as T1 and T2-weighted images are common imaging modalities at multiple PET and MRI centers with research programs in PD.

Overall, only 14 structural MRI studies (13 T1w, 1 T2w) did not use the PPMI dataset – and the average accuracy, sensitivity, and specificity for these studies was lower (on average, about 80% for each measure) compared to the PPMI cohort (on average, about 85% for each measure). While these differences were not statistically significant, there was a clear trend for better accuracy in PPMI datasets ($p=0.09$, 2-tailed t -test). This could be a result of better training of the models on the larger PPMI dataset but could also partly result from the influence of previously published literature on PPMI datasets on modeling choices made by newer investigations. Once again, it would be beneficial to challenge some of the best models trained on PPMI datasets to external data for structural imaging in PD.

Compared to structural MRI, only a few studies have been performed to investigate group discrimination in PD with ML/AI methods using resting state fMRI and DTI, and therefore, we recommend caution in the generalizability of these results. In this review, we included 12 studies with resting state-fMRI (rs-fMRI) data and only 9 with DTI. For rs-fMRI, SVM was the most common ML model used (8/12), along with individual reports of linear and logistic regression, canonical independent component analysis and a 3D-CNN, respectively. Overall, the accuracy of classification with rs-fMRI was similar to [^{18}F]FDG PET (about 80–85% across studies on average) with mixed results for SVM-based methods, but notably higher specificity was observed for rs-fMRI compared to [^{18}F]FDG PET (~85% for rs-fMRI vs. 70% for [^{18}F]FDG PET). It would also be of interest to compare the accuracy obtained using SVM (and other methods) in each study to the discrimination accuracy obtained through SSM-PCA method using fMRI. For DTI, only very few studies exist for a given ML/AI method (2 with SVM approaches, 2 with CNNs, 2 with radiomics or tract-based spatial statistics followed by radiomics, and some clustering and regression approaches). It is also important to note that the relatively modest accuracy of DTI-based studies (mean 79.2%) underscores important limitations of this modality in PD classification. This lower performance may be due to intrinsic constraints of limited effect sizes in microstructure imaging using diffusion imaging, limited sample sizes or both. It is important to note that most diffusion MRI studies have used DTI in PD, while more studies are needed with better sequences such as neurite orientation density and dispersion imaging or diffusion kurtosis imaging. It is also possible that while DTI is not an ideal modality for early detection of PD, it can still provide complementary microstructural alteration information that

could be more effective as part of a multimodal imaging framework to understand specific symptomology. Future studies integrating diffusion MRI with modalities such as structural MRI, dopaminergic SPECT and PET, and functional MRI may yield more mechanistic insights into different features of PD and related diseases. Further, about two-thirds of DTI AI/ML studies included relied on PPMI datasets, while only one-third of the fMRI studies reported used PPMI. There is room for more studies with these imaging modalities, and DTI ML/AI studies in particular, could benefit from exploring datasets outside PPMI. Further, it would be of interest if research groups who utilize rs-fMRI studies (especially using PPMI dataset), also apply the same (or similar) models to [^{18}F]FDG PET, including the SSM-PCA method as a benchmark.

This review also uncovered several limitations of the current applications of AI models to diagnostic classification tasks. First, there is in general an opaqueness in the reporting of and justification of feature selection methods. Studies varied widely in the approaches employed for this—ranging from manual selection of predefined regions of interest, to automated extraction of radiomics features, to high-dimensional data reduction methods such as PCA or LASSO. In many reports, details of feature selection were either limited or not described, which hampers reproducibility and complicates direct comparisons across studies. Given that feature selection strongly influences downstream model performance, future work should prioritize transparent reporting of these procedures and systematic evaluation of how different strategies could impact diagnostic accuracy. Second, while current research focuses mostly on improving performance metrics like high accuracy, addressing explainability in AI models has received little attention in this area of research. A suite of techniques has been developed, even for neuroimaging, which help peer inside the ‘black box’ of neural network models and shape our understanding of their inner working. This is extremely important to build confidence in AI models and understand conditions which might substantially affect performance, which is crucial for clinical translation. Future studies need to strike a better balance between reporting performance optimization and explainability. Third, a major limitation across modalities, particularly DaT SPECT, structural MRI, and DTI, is the heavy reliance on the PPMI dataset. While this resource has been invaluable in driving methodological development, the predominance of a single dataset raises concerns about overfitting and limits generalizability to more heterogeneous clinical populations. Very few studies have validated their models on external, independent datasets, leaving uncertainty about real-world applicability.

Future research should prioritize cross-cohort validation, inclusion of diverse demographic and clinical populations, and multi-site harmonization strategies to ensure that reported performance metrics translate into clinically robust tools. Authors should consider the use of standards developed for application of machine learning methods in clinical prediction models such as the TRIPOD + AI statement [146]. Lastly, we encourage readers to refer to a recent systematic review by Dzialis et al. [147] which evaluated similar limitations of neuroimaging-based AI studies in PD with further emphasis on issues pertaining to data splitting, data leakage, model complexity and performance reporting, all of which influence the methodological rigor and adoption of AI in clinical practice.

Conclusion

We performed a systematic review of the utility of ML/AI methods in discriminating PD from controls which included 136 studies across six different imaging modalities including SPECT, PET, structural MRI, fMRI and diffusion MRI. The best performance across sensitivity, specificity and accuracy was unsurprisingly obtained for CNN based methods in dopamine imaging. However, we also uncovered important trends in the performance results, which depended on algorithmic choices and data sources, across modalities. In particular, fMRI and [^{18}F]FDG PET showed good overall classification accuracy with SVM and other approaches, though more studies are needed with different ML/AI approaches for these modalities. We also noted a wider variability in the performance of specific classes of algorithms for structural MRI, which could either reflect overtraining of models in some studies or more susceptibility of structural MRI data to small choices in model parameters which needs greater scrutiny. Along with specific recommendations to improve confidence in the performance of ML/AI techniques for each modality, we also noted an emerging trend of over-reliance on one data source (PPMI) in some modalities (DaT SPECT, structural MRI, and DTI) which should be addressed by future researchers.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10072-025-08768-6>.

Author contributions Faranak Ebrahimian Sadabad conceptualized the review, designed the methodology, and led the writing of the manuscript. Praveen Honhar contributed to data analysis and manuscript writing. Shakiba Houshi, Seyed Faraz Nejati, Sara Bagherieh, Mahdie Hosseini, Fereshteh Yazdanpanah, and Salih Cayir participated in article screening and data extraction. Alexandria Brackett designed the search methodology in Covidence. David Matuskey and Hemant Tagare reviewed and approved the final manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability All data supporting the findings of this study are included within the article and its supplementary materials.

Declarations

Ethical approval Not applicable. This article is a review and does not contain any studies involving human participants or animals performed by any of the authors.

Conflict of interest The authors declare no conflicts of interest relevant to this manuscript.

References

- Willis AW, Roberts E, Beck JC, Fiske B, Ross W, Savica R et al (2022) Incidence of Parkinson disease in North America. *NPJ Parkinsons Dis* 8(1):170
- Chaudhuri KR, Azulay JP, Odin P, Lindvall S, Domingos J, Alobaidi A et al (2024) Economic burden of Parkinson's disease: a multinational, real-world, cost-of-illness study. *Drugs Real World Outcomes* 11(1):1–11
- Soman K, Nelson CA, Ceroni G, Goldman SM, Baranzini SE, Brown EG (2023) Early detection of Parkinson's disease through enriching the electronic health record using a biomedical knowledge graph. *Front Med (Lausanne)* 10:1081087
- Hughes AJDS, Ben-Shlomo Y, Lees AJ (2002) The accuracy of diagnosis of parkinsonian syndromes in a specialist movement disorder service. *Brain* 125(4):861–70
- Tolosa E, Garrido A, Scholz SW, Poewe W (2021) Challenges in the diagnosis of Parkinson's disease. *Lancet Neurol* 20(5):385–397
- Wang J, Xue L, Jiang J, Liu F, Wu P, Lu J et al (2024) Diagnostic performance of artificial intelligence-assisted PET imaging for parkinson's disease: a systematic review and meta-analysis. *NPJ Digit Med* 7(1):17
- Dennis AP, Strafella AP (2024) The role of AI and machine learning in the diagnosis of Parkinson's disease and atypical parkinsonisms. *Parkinsonism Relat Disord* 126:106986
- Beckers M, Bloem BR, Verbeek MM (2022) Mechanisms of peripheral levodopa resistance in Parkinson's disease. *NPJ Parkinsons Dis* 8(1):56
- Oren O, Gersh BJ, Bhatt DL (2020) Artificial intelligence in medical imaging: switching from radiographic pathological data to clinically meaningful endpoints. *Lancet Digit Health* 2(9):e486–e8
- Vitale A, Villa R, Ugga L, Romeo V, Stanzione A, Cuocolo R (2021) Artificial intelligence applied to neuroimaging data in parkinsonian syndromes: actuality and expectations. *Math Biosci Eng* 18(2):1753–1773
- Gabelloni M, Faggioni L, Borgheresi R, Restante G, Shortrede J, Tumminello L et al (2022) Bridging gaps between images and data: a systematic update on imaging biobanks. *Eur Radiol* 32(5):3173–3186
- Robert J, Gillies P, Paul E, Kinahan P, Hedvig Hricak M, PhD (2016) Dr(hc. gillies-et-al-2015-radiomics-images-are-more-than-pictures-they-are-data.pdf
- Giordano C, Brennan M, Mohamed B, Rashidi P, Modave F, Tighe P (2021) Accessing artificial intelligence for clinical decision-making. *Frontiers in Digital Health* 3:645232
- Khosravi M, Zare Z, Mojtabaiean SM, Izadi R (2024) Artificial intelligence and decision-making in healthcare: a thematic analysis of a systematic review of reviews. *Health Serv Res Manag Epidemiol* 11:23333928241234863
- Acosta JN, Falcone GJ, Rajpurkar P, Topol EJ (2022) Multimodal biomedical AI. *Nat Med* 28(9):1773–1784
- Vaillancourt DE, Barmpoutis A, Wu SS, DeSimone JC, Schauder M, Chen R et al (2025) Automated imaging differentiation for parkinsonism. *JAMA Neurol* 82(5):495–505
- Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372:n71
- McGowan J, Sampson M, Salzwedel DM, Cogo E, Foerster V, Lefebvre C (2016) Press peer review of electronic search strategies: 2015 guideline statement. *J Clin Epidemiol* 75:40–46
- Gionfriddo MR, McClendon C, Nolfi DA, Kalarchian MA, Covey JR (2024) Back to the basics: guidance for designing good literature searches. *Res Social Adm Pharm* 20(4):463–468
- Whiting PF, Rutjes AW, Westwood ME, Mallett S, Deeks JJ, Reitsma JB et al (2011) QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. *Ann Intern Med* 155(8):529–36
- Guido R, Ferrisi S, Lofaro D, Conforti D (2024) An overview on the advancements of support vector machine models in healthcare applications: a review. *Information* 15(4):235
- Murphy KP (2022) Probabilistic Machine Learning: An Introduction. The MIT Press, Cambridge, MA
- Avberšek LK, Repovš G (2022) Deep learning in neuroimaging data analysis: applications, challenges, and solutions. *Front Neuroimaging* 1:981642
- Krizhevsky A, Sutskever I, Hinton GE (2012) ImageNet classification with deep convolutional neural networks. *Proceedings of the 26th International Conference on Neural Information Processing Systems - Volume 1; Lake Tahoe, Nevada: Curran Associates Inc.; pp. 1097–105*
- Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*
- Huang G, Liu Z, Pleiss G, Maaten LV, Weinberger KQ (2022) Convolutional Networks with Dense Connectivity. *IEEE Trans Pattern Anal Mach Intell* 44(12):8704–8716. <https://doi.org/10.1109/TPAMI.2019.2918284>
- Bernal J, Kushibar K, Asfaw DS, Valverde S, Oliver A, Martí R et al (2019) Deep convolutional neural networks for brain image analysis on magnetic resonance imaging: a review. *Artif Intell Med* 95:64–81
- Chiu P-Y, Hou P-N, Hung G-U, Hsieh T-C, Chan P-K, Kao C-H (2022) Real-world testing of a machine learning-derived visual scale for Tc99m TRODAT-1 for diagnosing Lewy body disease: comparison with a traditional approach using semiquantification. *J Pers Med* 12(9):1369
- Yoon HJ, Cho K, Kim WG, Jeong Y-J, Jeong J-E, Kang D-Y (2021) Heterogeneity by global and textural feature analysis in F-18 FP-CIT brain PET images for diagnosis of Parkinson's disease. *Medicine* 100(35):e26961
- Magesh PR, Myloth RD, Tom RJ (2020) An explainable machine learning model for early detection of Parkinson's disease using LIME on DaTSCAN imagery. *Comput Biol Med* 126:104041
- Prashanth R, Roy SD, Mandal PK, Ghosh S (2016) High-accuracy detection of early parkinson's disease through multimodal features and machine learning. *Int J Med Inform* 90:13–21
- Bouchrouite J, Jilbab A, Jbari A (eds) (2022) Automatic SPECT Image Processing for Parkinson's Disease Early Detection. *International Conference on Smart Applications and Data Analysis*; Springer

33. Castillo-Barnes D, Merino-Chica J, Garcia-Diaz R, Jimenez-Mesa C, Arco JE, Ramirez J et al (eds) (2022) CAD system for Parkinson's disease with penalization of non-significant or high-variability input data sources. *International Work-Conference on the Interplay Between Natural and Artificial Computation*; : Springer
34. Bhalchandra NA, Prashanth R, Roy SD, Noronha S (eds) (2015) Early detection of Parkinson's disease through shape based features from 123 I-Ioflupane SPECT imaging. *IEEE 12th International Symposium on Biomedical Imaging (ISBI)*; 2015: IEEE
35. Khachnaoui H, Khlifa N, Mabrouk R (2022) Machine learning for early Parkinson's disease identification within SWEDD group using clinical and DaTSCAN SPECT imaging features. *J Imaging* 8(4):97
36. Kurmi A, Biswas S, Sen S, Sinitca A, Kaplun D, Sarkar R (2022) An ensemble of CNN models for Parkinson's disease detection using DaTscan images. *Diagnostics* 12(5):1173
37. Manimegalai R, Ramya A, Swedha S, Pruthvi M, Lokesh S (eds) (2022) Deep Learning Based Approach for Identification of Parkinson's Syndrome. *2022 International Conference on Intelligent Innovations in Engineering and Technology (ICIET)*; : IEEE
38. Martinez-Murcia FJ, Górriz JM, Ramírez J, Ortiz A (2018) Convolutional neural networks for neuroimaging in Parkinson's disease: is preprocessing needed? *Int J Neural Syst* 28(10):1850035
39. Oliveira FP, Castelo-Branco M (2015) Computer-aided diagnosis of Parkinson's disease based on [¹²³I] FP-CIT SPECT binding potential images, using the voxels-as-features approach and support vector machines. *J Neural Eng* 12(2):026008
40. Oliveira FP, Faria DB, Costa DC, Castelo-Branco M, Tavares JMR (2018) Extraction, selection and comparison of features for an effective automated computer-aided diagnosis of Parkinson's disease based on [¹²³I] FP-CIT SPECT images. *Eur J Nucl Med Mol Imaging* 45:1052–62
41. Pereira HR, Ferreira HA (eds) (2020) Classification of patients with Parkinson's disease using medical imaging and artificial intelligence algorithms. *XV Mediterranean Conference on Medical and Biological Engineering and Computing–MEDICON 2019: Proceedings of MEDICON 2019, September 26–28, 2019, Coimbra, Portugal*; : Springer
42. Pianpanit T, Lolak S, Sawangjai P, Sudhawiyangkul T, Wilaiprasitporn T (2021) Parkinson's disease recognition using SPECT image and interpretable AI: a tutorial. *IEEE Sens J* 21(20):22304–22316
43. Prashanth R, Roy SD, Mandal PK, Ghosh S (2016) High-accuracy classification of parkinson's disease through shape analysis and surface fitting in [¹²³I]-Ioflupane SPECT imaging. *IEEE J Biomed Health Inf* 21(3):794–802
44. Shiiba T, Arimura Y, Nagano M, Takahashi T, Takaki A (2020) Improvement of classification performance of Parkinson's disease using shape features for machine learning on dopamine transporter single photon emission computed tomography. *PLoS One* 15(1):e0228289
45. Shiiba T, Takano K, Takaki A, Suwazono S (2022) Dopamine transporter single-photon emission computed tomography-derived radiomics signature for detecting Parkinson's disease. *EJNMMI Res* 12(1):39
46. Thakur M, Kuresan H, Dhanalakshmi S, Lai KW, Wu X (2022) Soft attention based DenseNet model for Parkinson's disease classification using SPECT images. *Front Aging Neurosci* 14:908143
47. Wenzel M, Milletari F, Krüger J, Lange C, Schenk M, Apostolova I et al (2019) Automatic classification of dopamine transporter SPECT: deep convolutional neural networks can be trained to be robust with respect to variable image characteristics. *Eur J Nucl Med Mol Imaging* 46:2800–2811
48. Zhou Y, Tagare HD (eds) (2021) Self-normalized classification of Parkinson's disease DaTscan images. *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*; : IEEE
49. Choi H, Ha S, Im HJ, Paek SH, Lee DS (2017) Refining diagnosis of Parkinson's disease with deep learning-based interpretation of dopamine transporter imaging. *Neuroimage Clin* 16:586–594
50. Dehghan R, Naderan M, Alavi SE (eds) (2022) Detection of Parkinson's disease using Convolutional Neural Networks and Data Augmentation with SPECT images. *2022 12th International Conference on Computer and Knowledge Engineering (ICCKE)*; : IEEE
51. Ortiz A, Munilla J, Martínez-Ibañez M, Górriz JM, Ramírez J, Salas-Gonzalez D (2019) Parkinson's disease detection using iso-surfaces-based features and convolutional neural networks. *Front Neuroinform* 13:48
52. Quan J, Xu L, Xu R, Tong T, Su J (2019) DaTscan SPECT Image Classification for Parkinson's Disease. *arXiv preprint arXiv:190904142*
53. Castillo-Barnes D, Ramírez J, Segovia F, Martínez-Murcia FJ, Salas-Gonzalez D, Górriz JM (2018) Robust ensemble classification methodology for I123-Ioflupane SPECT images and multiple heterogeneous biomarkers in the diagnosis of Parkinson's disease. *Front Neuroinform* 12:53
54. Bollipo LM (2021) A novel supervised machine learning algorithm to detect Parkinson's disease on its early stages. *Turkish J Comput Math Educ (TURCOMAT)* 12(10):5257–5276
55. Jothi S, Anita S, Sivakumar S (2023) Modified exigent features block in jan net for analysing spect scan images to diagnose early-stage parkinson's disease. *Curr Med Imaging*. <https://doi.org/10.2174/1573405620666230605092654>
56. Khachnaoui H, Chikhaoui B, Khlifa N, Mabrouk R (2023) Enhanced Parkinson's disease diagnosis through convolutional neural network models applied to SPECT DaTSCAN images. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.3308075>
57. Hsu S-Y, Lin H-C, Chen T-B, Du W-C, Hsu Y-H, Wu Y-C et al (2019) Feasible classified models for Parkinson disease from 99mTc-TRODAT-1 SPECT imaging. *Sensors* 19(7):1740
58. Tagare HD, DeLorenzo C, Chelikani S, Saperstein L, Fulbright RK (2017) Voxel-based logistic analysis of PPMI control and parkinson's disease DaTscans. *Neuroimage* 152:299–311
59. Glaab E, Trezzi J-P, Greuel A, Jäger C, Hodak Z, Drzezga A et al (2019) Integrative analysis of blood metabolomics and PET brain neuroimaging data for parkinson's disease. *Neurobiol Dis* 124:555–562
60. Shen T, Jiang J, Lin W, Ge J, Wu P, Zhou Y et al (2019) Use of overlapping group LASSO sparse deep belief network to discriminate parkinson's disease and normal control. *Front NeuroSci* 13:396
61. Manzanera OM, Meles SK, Leenders KL, Renken RJ, Paganì M, Arnaldi D et al (2019) Scaled subprofile modeling and convolutional neural networks for the identification of Parkinson's disease in 3D nuclear imaging data. *Int J Neural Syst* 29(09):1950010
62. Sun X, Ge J, Li L, Zhang Q, Lin W, Chen Y et al (2022) Use of deep learning-based radiomics to differentiate Parkinson's disease patients from normal controls: a study based on [18F] FDG PET imaging. *Eur Radiol* 32(11):8008–18
63. van Veen R, Talavera Martinez L, Kogan RV, Meles SK, Mudali D, Roerdink JB et al (2018) Machine learning based analysis of FDG-PET image data for the diagnosis of neurodegenerative diseases. *IOS, Applications of Intelligent Systems*, pp 280–289
64. Wu Y, Jiang J-H, Chen L, Lu J-Y, Ge J-J, Liu F-T et al (2019) Use of radiomic features and support vector machine to distinguish Parkinson's disease cases from normal controls. *Ann Transl Med*. <https://doi.org/10.21037/atm.2019.11.26>

65. Ruppert-Junck MC, Kräling G, Greuel A, Tittgemeyer M, Timmermann L, Drzezga A et al (2024) Random forest analysis of midbrain hypometabolism using [18F]-FDG PET identifies Parkinson's disease at the subject-level. *Front Comput Neurosci* 18:1328699
66. Dai Y, Song Y, Liu W, Bai W, Gao Y, Dong X et al (2021) Multi-focus image fusion based on Convolution neural network for parkinson's disease image classification. *Diagnostics* 11(12):2379
67. Iep A, Chawki MB, Goldfarb L, Nguyen L, Brulon V, Comtat C et al (2023) Relevance of 18F-DOPA visual and semi-quantitative PET metrics for the diagnostic of Parkinson disease in clinical practice: a machine learning-based inference study. *EJNMMI Res* 13(1):13
68. Martins R, Oliveira F, Moreira F, Moreira AP, Abrunhosa A, Januário C et al (2021) Automatic classification of idiopathic parkinson's disease and atypical parkinsonian syndromes combining [11 C] Raclopride PET uptake and MRI grey matter morphometry. *J Neural Eng* 18(4):046037
69. Piccardo A, Cappuccio R, Bottoni G, Cecchin D, Mazzella L, Cirone A et al (2021) The role of the deep convolutional neural network as an aid to interpreting brain [18 F] DOPA PET/CT in the diagnosis of parkinson's disease. *Eur Radiol* 31:7003–7011
70. Salvatore C, Cerasa A, Castiglioni I, Gallivanone F, Augimeri A, Lopez M et al (2014) Machine learning on brain MRI data for differential diagnosis of Parkinson's disease and Progressive Supranuclear Palsy. *J Neurosci Methods* 222:230–7
71. Focke NK, Helms G, Scheewe S, Pantel PM, Bachmann CG, Dechent P et al (2011) Individual voxel-based subtype prediction can differentiate progressive supranuclear palsy from idiopathic Parkinson syndrome and healthy controls. *Hum Brain Mapp* 32(11):1905–15
72. Wang Y, He N, Zhang C, Zhang Y, Wang C, Huang P et al (2023) An automatic interpretable deep learning pipeline for accurate Parkinson's disease diagnosis using quantitative susceptibility mapping and T1-weighted images. *Wiley Online Library*, pp 1065–9471. Report No
73. Kiryu S, Yasaka K, Akai H, Nakata Y, Sugomori Y, Hara S et al (2019) Deep learning to differentiate parkinsonian disorders separately using single midsagittal MR imaging: a proof of concept study. *Eur Radiol* 29:6891–9
74. Lampe L, Niehaus S, Huppertz H-J, Merola A, Reinelt J, Mueller K et al (2022) Comparative analysis of machine learning algorithms for multi-syndrome classification of neurodegenerative syndromes. *Alzheimers Res Ther* 14(1):62
75. Lee P-L, Chou K-H, Lu C-H, Chen H-L, Tsai N-W, Hsu A-L et al (2018) Extraction of large-scale structural covariance networks from grey matter volume for parkinson's disease classification. *Eur Radiol* 28:3296–3305
76. Tupe-Waghmare P, Rajan A, Prasad S, Saini J, Pal PK, Ingalhalikar M (2021) Radiomics on routine T1-weighted MRI can delineate parkinson's disease from multiple system atrophy and progressive supranuclear palsy. *Eur Radiol* 31(11):8218–8227
77. Rana B, Juneja A, Saxena M, Gudwani S, Kumaran SS, Agrawal R et al (eds) (2016) Voxel-based morphometry and minimum redundancy maximum relevance method for classification of Parkinson's disease and controls from T1-weighted MRI. *Proceedings of the Tenth Indian Conference on Computer Vision, Graphics and Image Processing*
78. Duchesne S, Rolland Y, Verin M (2009) Automated computer differential classification in parkinsonian syndromes via pattern analysis on MRI. *Acad Radiol* 16(1):61–70
79. Huppertz HJ, Möller L, Südmeyer M, Hilker R, Hattingen E, Egger K et al (2016) Differentiation of neurodegenerative parkinsonian syndromes by volumetric magnetic resonance imaging analysis and support vector machine classification. *Mov Disord* 31(10):1506–1517
80. Gore S, Dhole A, Kumbhar S, Jagtap J, Thakare A, Gudadhe S (2023) Radiomics for Parkinson's disease classification using advanced texture-based biomarkers. *MethodsX* 11:102359
81. Cigdem O, Demirel H (2018) Performance analysis of different classification algorithms using different feature selection methods on Parkinson's disease detection. *J Neurosci Methods* 309:81–90
82. Peng B, Wang S, Zhou Z, Liu Y, Tong B, Zhang T et al (2017) A multilevel-ROI-features-based machine learning method for detection of morphometric biomarkers in parkinson's disease. *Neurosci Lett* 651:88–94
83. Singh G, Vadera M, Samavedham L, Lim EC-H (2019) Multiclass diagnosis of neurodegenerative diseases: a neuroimaging machine-learning-based approach. *Ind Eng Chem Res* 58(26):11498–11505
84. Sun D, Wu X, Xia Y, Wu F, Geng Y, Zhong W et al (2021) Differentiating Parkinson's disease motor subtypes: a radiomics analysis based on deep gray nuclear lesion and white matter. *Neurosci Lett* 760:136083
85. Amoroso N, La Rocca M, Monaco A, Bellotti R, Tangaro S (2018) Complex networks reveal early MRI markers of Parkinson's disease. *Med Image Anal* 48:12–24
86. Yang M, Huang X, Huang L, Cai G (2023) Diagnosis of Parkinson's disease based on 3D ResNet: the frontal lobe is crucial. *Biomed Signal Process Control* 85:104904
87. Singh G, Samavedham L (2015) Unsupervised learning based feature extraction for differential diagnosis of neurodegenerative diseases: a case study on early-stage diagnosis of Parkinson disease. *J Neurosci Methods* 256:30–40
88. Chakraborty S, Aich S, Kim H-C (2020) Detection of Parkinson's disease from 3T T1 weighted MRI scans using 3D convolutional neural network. *Diagnostics* 10(6):402
89. Chattopadhyay T, Singh A, Laltoo E, Boyle CP, Owens-Walton C, Chen Y-L et al (eds) (2023) Comparison of Anatomical and Diffusion MRI for detecting Parkinson's Disease using Deep Convolutional Neural Network. 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC); : IEEE
90. Dhinagar NJ, Thomopoulos SI, Owens-Walton C, Stripelis D, Ambite JL, Ver Steeg G et al (eds) (2021) 3D convolutional neural networks for classification of Alzheimer's and Parkinson's disease with T1-weighted brain MRI. 17th International Symposium on Medical Information Processing and Analysis; : SPIE
91. Tafuri B, Lombardi A, Nigro S, Urso D, Monaco A, Pantaleo E et al (2022) The impact of harmonization on radiomic features in Parkinson's disease and healthy controls: a multicenter study. *Front Neurosci* 16:1012287
92. Ganotra R, Gupta S (2023) Detection of Parkinson's Disease Using Support Vector Machine and Combination of Various Tissue Density Features. *Recent Developments in Electrical and Electronics Engineering: Select Proceedings of ICRDEEE 2022*: Springer; pp. 65–9
93. Pahuja G, Nagabhushan T, Prasad B, Pushkarna R (2018) Early detection of Parkinson's disease through multimodal features using machine learning approaches. *Int J Signal Imaging Syst Eng* 11(1):31–43
94. Song C, Zhao W, Jiang H, Liu X, Duan Y, Yu X et al (2021) Stability evaluation of brain changes in parkinson's disease based on machine learning. *Front Comput Neurosci* 15:735991
95. Zeng L-L, Xie L, Shen H, Luo Z, Fang P, Hou Y et al (2017) Differentiating patients with parkinson's disease from normal controls using Gray matter in the cerebellum. *Cerebellum* 16(1):151–157
96. Erdaş ÇB, Sümer E (2023) A fully automated approach involving neuroimaging and deep learning for Parkinson's disease detection and severity prediction. *PeerJ Computer Science* 9:e1485

97. Camacho M, Wilms M, Mouches P, Almgren H, Souza R, Camicioli R et al (2023) Explainable classification of Parkinson's disease using deep learning trained on a large multi-center database of T1-weighted MRI datasets. *Neuroimage Clin* 38:103405
98. Li C, Hui D, Wu F, Xia Y, Shi F, Yang M et al (2024) Automatic diagnosis of Parkinson's disease using artificial intelligence base on routine T1-weighted MRI. *Front Med (Lausanne)* 10:1303501
99. Babu GS, Suresh S, Mahanand BS (2014) A novel PBL-McRBFN-RFE approach for identification of critical brain regions responsible for parkinson's disease. *Expert Syst Appl* 41(2):478–488
100. Chen Y, Storrs J, Tan L, Mazlack LJ, Lee J-H, Lu LJ (2014) Detecting brain structural changes as biomarker from magnetic resonance images using a local feature based SVM approach. *J Neurosci Methods* 221:22–31
101. Solana-Lavalle G, Rosas-Romero R (2021) Classification of PPMI MRI scans with voxel-based morphometry and machine learning to assist in the diagnosis of Parkinson's disease. *Comput Methods Programs Biomed* 198:105793
102. Yang Y, Wei L, Hu Y, Wu Y, Hu L, Nie S (2021) Classification of Parkinson's disease based on multi-modal features and stacking ensemble learning. *J Neurosci Methods* 350:109019
103. Sivaranjini S, Sujatha C (2024) Analysis of cognitive dysfunction in Parkinson's disease using voxel based morphometry and radiomics. *Cogn Process*. <https://doi.org/10.1007/s10339-024-01197-x>
104. Kaur S, Aggarwal H, Rani R (2021) Diagnosis of parkinson's disease using deep CNN with transfer learning and data augmentation. *Multimedia Tools Appl* 80(7):10113–10139
105. Shi J, Xue Z, Dai Y, Peng B, Dong Y, Zhang Q et al (2018) Cascaded multi-column RVFL+classifier for single-modal neuroimaging-based diagnosis of parkinson's disease. *IEEE Trans Biomed Eng* 66(8):2362–2371
106. Esmailzadeh S, Yang Y, Adeli E (2018) End-to-end Parkinson disease diagnosis using brain MR-images by 3D-CNN. *arXiv preprint arXiv:1806.05233*
107. Gong B, Shi J, Ying S, Dai Y, Zhang Q, Dong Y et al (2018) Neuroimaging-based diagnosis of parkinson's disease with deep neural mapping large margin distribution machine. *Neurocomputing* 320:141–149
108. Adeli E, Shi F, An L, Wee C-Y, Wu G, Wang T et al (2016) Joint feature-sample selection and robust diagnosis of parkinson's disease from MRI data. *NeuroImage* 141:206–219
109. Praneeth P, Sathvika M, Kommareddy V, Sarath M, Mallela ST, Vani KS et al (2023) Classification of parkinson's disease in brain MRI images using deep residual convolutional neural network (DRCNN). *Int J Comput Inform Syst Industrial Manage Appl* 15:13
110. Begum R, Kumar TP, Rao MRN (2023) Deep convolutional neural networks for diagnosis of Parkinson's disease using MRI data. *Ingénierie des Systèmes d'Information*. <https://doi.org/10.18280/isi.280324>
111. Kanagaraj S, Hema M, Nageswara Guptha M (2024) An improved approach for early diagnosis of Parkinson's disease using advanced DL models and image alignment. *Automatika: časopis za automatiku, mjerenje, elektroniku, računarstvo i komunikacije* 65(3):911–24
112. Sailaja B, VenuGopal T (2023) Brain MRI image classification and analysis using modified ResNet50V2 for Parkinson's disease detection. *SN Comput Sci* 4(6):854
113. Yasaka K, Kamagata K, Ogawa T, Hatano T, Takeshige-Amano H, Ogaki K et al (2021) Parkinson's disease: deep learning with a parameter-weighted structural connectome matrix for diagnosis and neural circuit disorder investigation. *Neuroradiology*. <https://doi.org/10.1007/s00234-021-02648-4>
114. Chen Y, Zhu G, Liu D, Liu Y, Yuan T, Zhang X et al (2020) The morphology of thalamic subnuclei in Parkinson's disease and the effects of machine learning on disease diagnosis and clinical evaluation. *J Neurol Sci* 411:116721
115. Cui X, Zhou Y, Zhao C, Li J, Zheng X, Li X et al (2023) A multiscale hybrid attention networks based on multiview images for the diagnosis of parkinson's disease. *IEEE Trans Instrum Meas* 73:1–11
116. Peng B, Gong Z, Zhang Y, Shen B, Pang C, Zhang L et al (2022) Self-paced learning and privileged information based KRR classification algorithm for diagnosis of parkinson's disease. *Neurosci Lett* 766:136312
117. Vyas T, Yadav R, Solanki C, Darji R, Desai S, Tanwar S (2022) Deep learning-based scheme to diagnose Parkinson's disease. *Expert Syst* 39(3):e12739
118. Sivaranjini S, Sujatha CM (2020) Deep learning based diagnosis of Parkinson's disease using convolutional neural network. *Multimedia Tools Appl* 79(21):15467–15479
119. Yin D, Zhao Y, Wang Y, Zhao W, Hu X (2020) Auxiliary diagnosis of heterogeneous data of parkinson's disease based on improved convolution neural network. *Multimedia Tools Appl* 79:24199–24224
120. Liu P, Wang H, Zheng S, Zhang F, Zhang X (2020) Parkinson's disease diagnosis using neostriatum radiomic features based on T2-weighted magnetic resonance imaging. *Front Neurol* 11:248
121. Hussain SS, Degang X, Shah PM, Islam SU, Alam M, Khan IA et al (2023) Classification of Parkinson's disease in patch-based MRI of substantia nigra. *Diagnostics* 13(17):2827
122. Akram SW, Kumar AS (2024) Parkinson's disease diagnosis from T1 and T2 weighted magnetic resonance images using FBLstmNet architecture. *Multimedia Tools Appl* 83(19):56837–56864
123. Veetil IK, Gopalakrishnan E, Sowmya V, Soman K (eds) (2021) Parkinson's disease classification from magnetic resonance images (MRI) using deep transfer learned convolutional neural networks. 2021 IEEE 18th India Council International Conference (INDICON); : IEEE
124. Tian Z-y, Qian L, Fang L, Peng X-h, Zhu X-h, Wu M et al (2020) Frequency-specific changes of resting brain activity in Parkinson's disease: a machine learning approach. *Neuroscience* 436:170–83
125. Long D, Wang J, Xuan M, Gu Q, Xu X, Kong D et al (2012) Automatic classification of early parkinson's disease with multimodal MR imaging. *PLoS ONE* 7(11):e47714
126. Rubbert C, Mathys C, Jockwitz C, Hartmann CJ, Eickhoff SB, Hoffstaedter F et al (2019) Machine-learning identifies parkinson's disease patients based on resting-state between-network functional connectivity. *Br J Radiol* 92(1101):20180886
127. Shi D, Yao X, Li Y, Zhang H, Wang G, Wang S et al (2022) Classification of Parkinson's disease using a region-of-interest and resting-state functional magnetic resonance imaging-based radiomics approach. *Brain Imaging Behav* 16(5):2150–63
128. Shi D, Zhang H, Wang G, Wang S, Yao X, Li Y et al (2022) Machine learning for detecting parkinson's disease by resting-state functional magnetic resonance imaging: a multicenter radiomics analysis. *Front Aging Neurosci* 14:806828
129. Wu J, Ma L, Luo D, Jin Z, Wang L, Wang L et al (2024) Functional and structural gradients reveal atypical hierarchical organization of Parkinson's disease. *Hum Brain Mapp* 45(4):e26647
130. Islam NU, Khanam R, Kumar A (2023) Using 3D CNN for classification of Parkinson's disease from resting-state fMRI data. *J Eng Appl Sci* 70(1):89
131. Chen B, Cui W, Wang S, Sun A, Yu H, Liu Y et al (2023) Functional connectome automatically differentiates multiple system atrophy (parkinsonian type) from idiopathic parkinson's disease at early stages. *Hum Brain Mapp* 44(6):2176–2190
132. Xu N, Zhou Y, Patel A, Zhang N, Liu Y (2023) Parkinson's disease diagnosis beyond clinical features: a bio-marker using

- topological machine learning of resting-state functional magnetic resonance imaging. *Neuroscience* 509:43–50
133. Dutta SB, Vig R (2024) Performance evaluation of dictionary learning and ICA on parkinson's patients classification using machine learning. *Multimedia Tools Appl* 83(8):24467–24483
 134. Abós A, Baggio HC, Segura B, García-Díaz AI, Compta Y, Martí MJ et al (2017) Discriminating cognitive status in Parkinson's disease through functional connectomics and machine learning. *Sci Rep* 7(1):45347
 135. Cao X, Wang X, Xue C, Zhang S, Huang Q, Liu W (2020) A radiomics approach to predicting parkinson's disease by incorporating whole-brain functional activity and gray matter structure. *Front Neurosci* 14:751
 136. Tsai C-C, Chen Y-L, Lu C-S, Cheng J-S, Weng Y-H, Lin S-H et al (2023) Diffusion tensor imaging for the differential diagnosis of Parkinsonism by machine learning. *Biomed J* 46(3):100541
 137. Li J, Liu X, Wang X, Liu H, Lin Z, Xiong N (2022) Diffusion tensor imaging radiomics for diagnosis of Parkinson's disease. *Brain Sci* 12(7):851
 138. Zhao H, Tsai C-C, Zhou M, Liu Y, Chen Y-L, Huang F et al (2022) Deep learning based diagnosis of Parkinson's disease using diffusion magnetic resonance imaging. *Brain Imaging Behav* 16(4):1749–60
 139. Huang Z, Lei H, Chen G, Frangi AF, Xu Y, Elazab A et al (2021) Parkinson's disease classification and clinical score regression via united embedding and sparse learning from longitudinal data. *IEEE Trans Neural Networks Learn Syst* 33(8):3357–3371
 140. Zhang Q, Wang H, Shi Y, Li W (2024) White matter biomarker for predicting de novo Parkinson's disease using tract-based spatial statistics: a machine learning-based model. *Quant Imaging Med Surg* 14(4):3086
 141. Lei H, Huang Z, Zhou F, Elazab A, Tan EL, Li H et al (2019) Parkinson's disease diagnosis via joint learning from multiple modalities and relations. *IEEE J* 23(4):1437–1449
 142. Prasuhn J, Heldmann M, Münte TF, Brüggemann N (2020) A machine learning-based classification approach on Parkinson's disease diffusion tensor imaging datasets. *Neurol Res Pract* 2:1–5
 143. Zhao H, Tsai C-C, Zhu C, Zhou M, Wang J-J, Liu Y (eds) (2021) *Deep Learning-Based Regional Sub-models Integration for Parkinson's Disease Diagnosis Using Diffusion Tensor Imaging*. International Conference on Image and Graphics; : Springer.
 144. Wakasugi N, Takano H, Abe M, Sawamoto N, Murai T, Mizuno T et al (2024) Harmonizing multisite data with the ComBat method for enhanced Parkinson's disease diagnosis via DAT-SPECT. *Front Neurol* 15:1306546
 145. Spetsieris PG, Ma Y, Dhawan V, Eidelberg D (2009) Differential diagnosis of parkinsonian syndromes using PCA-based functional imaging features. *Neuroimage* 45(4):1241–1252
 146. Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B et al (2024) TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 385:e078378
 147. Dzialas V, Doering E, Eich H, Strafella AP, Vaillancourt DE, Simonyan K et al (2024) Houston, we have AI problem! Quality issues with neuroimaging-based artificial intelligence in Parkinson's disease: a systematic review. *Mov Disord* 39(12):2130–43

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